

Mohammad Rafeian

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RESEARCH INTERESTS

Program analysis, automated software testing, and software security — applying machine learning and rigorous empirical methods to improve the soundness and precision of static analysis and call-graph tools.

EDUCATION

University of Texas at Dallas, M.S. & Ph.D. in Software Engineering & Security 2023 – Present
M.S. May 2026 · Ph.D. expected May 2027 · Advisor: Dr. Shiyi Wei

Isfahan University, B.S. in Computer Engineering Aug 2019 – Jun 2023
ACM Student Chapter Member

PUBLICATIONS

1. “Is Call Graph Pruning Really Effective? An Empirical Re-evaluation.” M. Rafeian, V. Birsan, K. Katiyar, D. Zhong, S. Wei. *ICSE 2026*.
2. “Improving ML-based Static Analysis Classification via Explainable AI.” S. Yerramreddy, M. Rafeian, S. Wei, A. Porter. *ICST 2026*.
3. “Agentic Root Cause Reasoning and Unsoundness in Call Graph Analysis.” M. Rafeian, M. Hicks, S. Wei. *ICSE 2027 (under review)*.
4. “Automated White-Box Unit Testing for Functional Requirements.” Y. Zhou, M. Rafeian, S. Wei, W. Lam, A. Marcus. *ICSE 2027 (under review)*.
5. “An Empirical Study of Static Analysis-Based Variability Bug Detection.” A. Mordahl, Z. Patterson, M. Rafeian, M. T. Ahmad, S. Wei. *ESEM 2026 (under review)*.
6. “Automatic Test Suites for Static Analysis Tools via Dynamic Analysis.” A. Mordahl, M. Rafeian, S. Wei. *FSE 2027 (in submission)*.

RESEARCH EXPERIENCE

Graduate Research Assistant — University of Texas at Dallas Aug 2023 – Present
Systems Software & Security Lab · Advisor: Dr. Shiyi Wei

- Lead research in program analysis and security, applying machine learning to automated vulnerability detection and testing for C/Java systems.
- Designed large-scale, reproducible experimental pipelines using fuzzing and dynamic bytecode instrumentation to benchmark four static analysis tools.
- Introduced three novel datasets and evaluation methodologies for static analysis, resulting in six papers across ICSE, ICST, ESEM, and FSE.

Research Assistant — Model-Driven SE Lab, Isfahan University Jan 2022 – Jan 2023

- Built an automatic reservation-system generator engine (MDSE) and co-authored a journal paper on automatically generating full web applications.

SELECTED RESEARCH PROJECTS

Call Graph Pruning Framework · Python, Java, Bash Feb – Jul 2025

- Built static call graphs (DOOP, OPAL, WALA) and instrumented Java bytecode (ASM); fine-tuned CodeBERT/CodeT5 to improve edge classification by 30%.
- Re-evaluated prior pruning methods and fixed dataset flaws, showing reported gains were inflated (+50% claimed vs. +5% actual).

Explainable AI for Static Analysis · Python, Java, C, Bash Jan – Sep 2025

- Constructed the first explainable dataset for static-analysis ML (1,132 Java + 67K C samples), evaluating 7 code LLMs.
- Applied Explanation-Guided Learning to raise relevant-line focus from 35% to 65% and improve classification accuracy by 20%.

Fuzzing for Static Analysis Benchmarking · Python, Java, C Jul 2025 – Present

- Built a hybrid (directed + grey-box) fuzzing pipeline with a novel feedback mechanism that auto-generates static-analysis benchmarks, saving ~80 person-hours.

ACADEMIC SERVICE

- **Artifact Evaluation Committee** — PLDI, ASE, ISSTA, and ICST (2026)

TECHNICAL SKILLS

Program Analysis: WALA, DOOP, Soot, OPAL, Jazzer, pointer & call-graph analysis, bytecode instrumentation (ASM)

Machine Learning: PyTorch, TensorFlow, scikit-learn, LLM fine-tuning (CodeBERT/CodeT5), Explainable AI

Languages & Tools: Python, Java, C/C++, Scala, JavaScript, SQL · Git, Docker, Linux, PostgreSQL, Redis